

Gas Absorption in a Packed Bed

Pre-Lab Plan

Unit Operations Lab # 

Date: 1/22/2020

Group 3, Thursday, AM section

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1. Experimental Objective (5 pts)

In this lab, the objective is to determine fluid flow characteristics, gas absorption characteristics, flooding point, and overall mass transfer coefficient by using a packed bed gas absorber to remove CO_2 with water in an air stream.

2. Experimental Apparatus (5 pts)

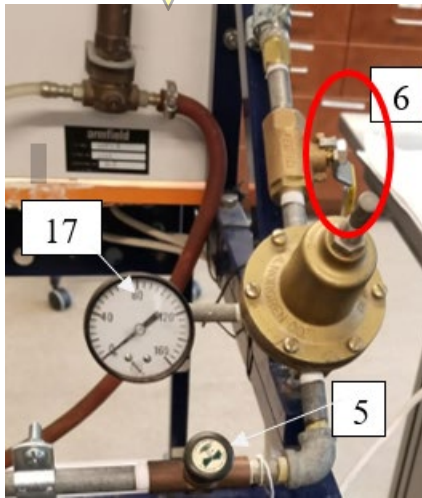


Figure 1: Air Supply

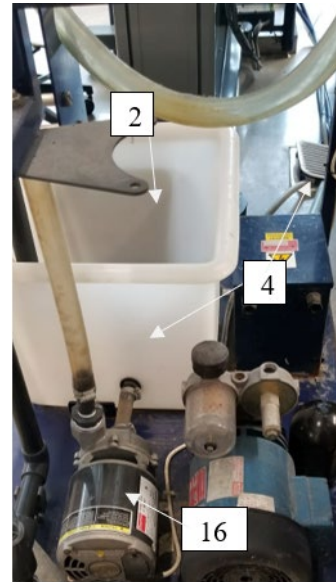


Figure 2: Lower Apparatus

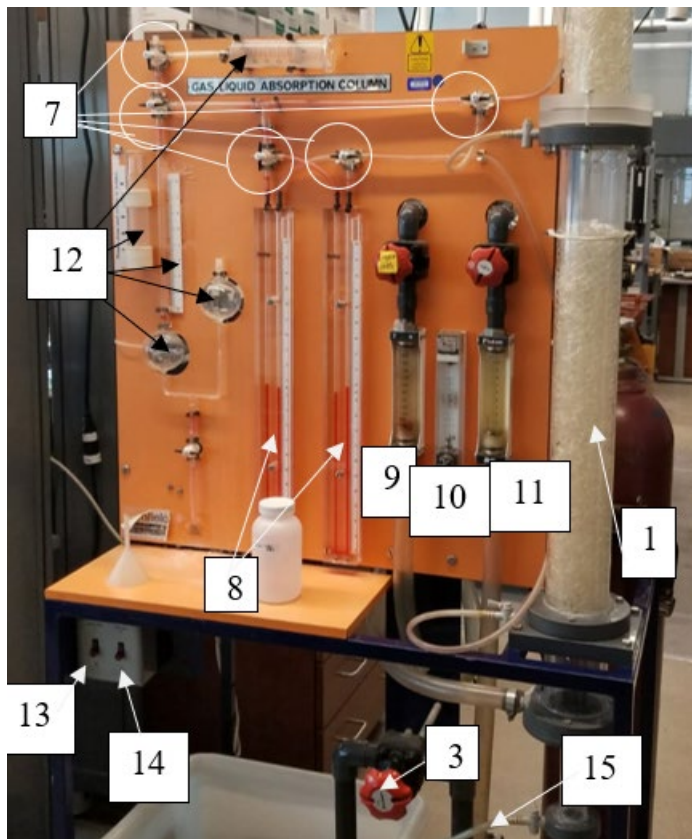


Figure 3: Gas Absorption

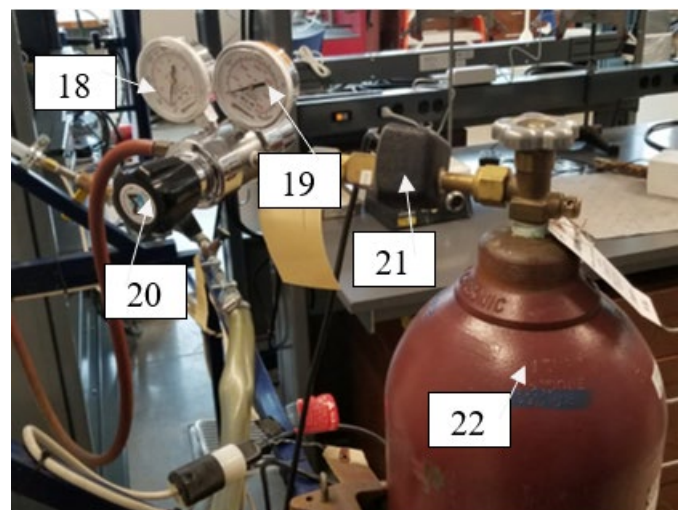


Figure 4: CO_2 Supply

Table of Equipment

Number	Name of Equipment	Description
1	Gas absorption column	Holds the gas and liquid components and where CO_2 is separated
2	Liquid reservoir tank	Holds de-ionized water from tap
3	Control valve for reservoir tank	Controls flowrate of water into column
4	Drainage valve (located within the reservoir)	Drains collected liquid from reservoir
5	Control valve for air	Controls flowrate of air into column
6	Air shut off valve	Opens/closes line from compressed air
7	Manometer valves	Controls the flow of gas mixture to manometers and Hempl apparatus
8	Manometers	Measures pressure drop across the column
9	Air rotameter	Measures air flowrate
10	Carbon dioxide rotameter	Measures carbon dioxide flowrate
11	Water rotameter	Measures water flowrate
12	Hempl apparatus (piston, cylinder, container bulbs, level measure)	Takes sample of gas mixture volume and determines the concentration of CO_2
13	Water pump switch	Sends power to water pump
14	CO_2 heater switch	Heats up CO_2 line from cylinder
15	Sample valve	Takes water (containing CO_2) sample from column
16	Main water pump	Pumps water to column
17	Air pressure gauge	Measures pressure of compressed air (PSI)
18	Supply pressure gauge	Measures pressure of CO_2 going to column
19	Cylinder pressure gauge	Measures pressure of CO_2 left in cylinder
20	Supply pressure valve	Changes the flowrate of CO_2 to column
21	CO_2 heating element	Heats the supply of CO_2 from cylinder
22	CO_2 cylinder	Stores CO_2 at high pressure
23	Supply needle valve	Fine-tunes the flowrate of CO_2 to column

2.2 Materials and Supplies (5 pts)

Table of Material/Supplies	
Name of Equipment	Purpose
Buret/buret clamp (50mL)	Contains standard base for titration of CO_2 -absorbed water
Graduated cylinder (50mL)	Used to collect sample
Stopwatch	For intervals of samples
1-50mL Pipet	Drops indicator for Titration
125mL Erlenmeyer Flasks	Titration vessel holding sample
Thermometer	Measures temperature of water in reservoir
Compressed Air	Supplied to column
Phenolphthalein Indicator Solution	Titration indicator
0.05N or 0.01N Sodium Hydroxide Solution	Used in Hempl apparatus to determine CO_2 concentration
CO_2 gas	Supplied to column
Deionized Water	Supplied to column from reservoir
1.0N Sodium Hydroxide Solution	Titration standard base for CO_2 -absorbed water

2.3 Experimental Plan (10 pts)

Determine the pressure drop across the packed bed(1A):

- 1) Close valve C4 and valve D1
- 2) Use deionized water and fill the liquid reservoir tank to $\frac{3}{4}$ full
- 3) Record Temperature of the deionized water
- 4) Change the flowrate of F2 to 20 l/min using valve C2
- 5) Wait 5 minutes for air to the column then measure the pressure drop\
- 6) Open valve C4, close valve C2 and C3 and start the liquid pump and open valve C1
- 7) Turn valve C1 for 15-30 seconds to remove air from the pump
- 8) Adjust the flow of water to 1.4 l/min on F1 by adjusting C1
- 9) Open the gas flow control valve C2 and just F2 to 20 l/min
- 10) Measure the pressure drop as you increase the flowrate until flooding occurs
- 11) Repeat the experiment for 3,6,8 l/min

Bringing Water to equilibrium with CO_2 and analysis of CO_2 in water(2A):

- 12) Follow steps 1-3 from 1A and collect a sample and determine the initial CO_2 concentration.
- 13) Open valve C4, close valve C2 and C3 and start the liquid pump and open valve C1
- 14) Turn valve C1 for 15-30 seconds to remove air from the pump
- 15) Adjust the flow of water to 6 l/min on F1 by adjusting C1
- 16) Open the gas flow control valve C2 and adjust F2 to 20 l/min

17) Open the main valve from the Carbon Dioxide Cylinder and then adjust the Carbon Dioxide regulator to 20 Psi

18) Adjust valve C3 so that F3 is between 2-5 l/min

19) After 10 minutes of operation take 50 mL samples from S4 until the CO₂ values are constant

20) Measure the volume fraction of CO₂ leaving the column from S2 and stop once the CO₂ levels are constant

Desorption of CO₂ into laboratory air(2b):

21) Shut off CO₂ (Valve C3)

22) Decrease the liquid flowrate to 1.5 l/min

23) Open the drain valve D1 and close valve C4

24) Adjust D1 to maintain a liquid level above S4 and Take samples in 1-minute intervals

25) Analyze the concentration

Absorption of CO₂ by H₂O in a continuous column mode(3):

26) Follow steps 1-3 from 1A and collect a sample and determine the initial CO₂ concentration

27) Open valve C4, close valve C2 and C3 and start the liquid pump and open valve C1

28) Turn valve C1 for 15-30 seconds to remove air from the pump

29) Open D1 and close valve C4 and use D1 To maintain a liquid level below the air-CO₂ inlet but above the sampling valve S4 and leave S4 open and put bucket underneath

30) Adjust F2 to 20 l/min and open valve on the carbon dioxide cylinder

31) Adjust the regulator to 20 Psi

32) Adjust valve C3 so that F3 is between 2-6 l/min

33) Immediately take 50 mL samples from S4 until the CO₂ values are constant every 30 seconds for two minutes.

34) Then take a sample every 1 minute for 5 minutes and then take a sample every five minutes until constant

35) Analyze gas samples from S2(Appendix A)

Project Safety Evaluation (10 pts)

Project Title: Gas Absorption_____

Date: 1/23/2020

	Yes/No	
Potential electrical hazards?	Yes	If yes, identify conditions: Heater heats up the CO ₂ , otherwise the CO ₂ would freeze the pipes Power to pump
Potential mechanical hazards?	YES	If yes, identify conditions: Pump for water
Potential pressure hazards?	YES	If yes, identify conditions: There is a compressed CO ₂ cylinder and compressed air line

